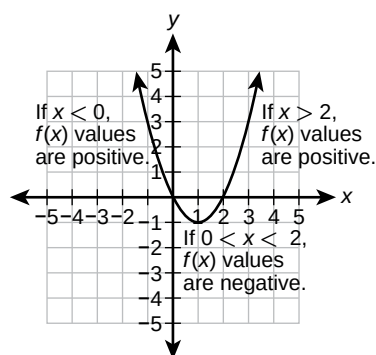


Working with Function Models

Relative maximum and minimum points; increasing, decreasing, and constant intervals; and classification as even, odd, or neither are important features of functions.

The graph of a function can help you get an understanding of the relationships between two quantities. For example, the graph of the function $f(x) = x^2 - 2x$ shows that as the x -values increase, the values of $f(x)$ change from being positive values to being negative values and then return to positive values.



Finding Relative Maximum and Minimum Points

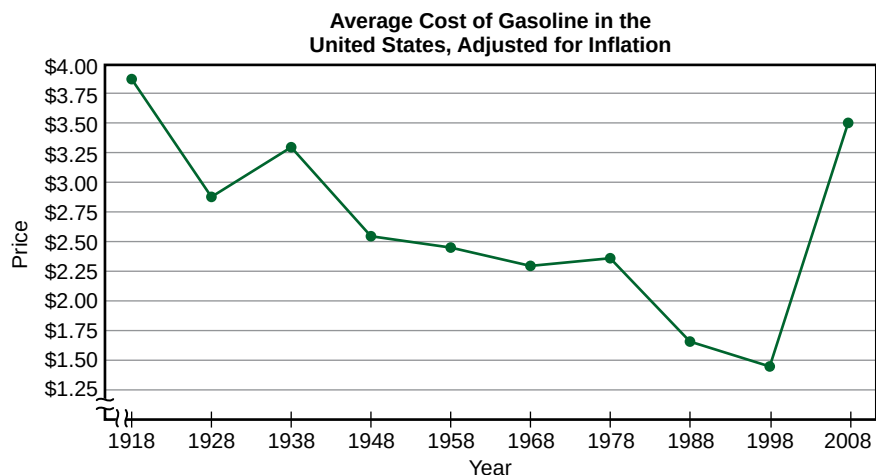
Definitions

A **relative maximum point** of a function is a point that has a greater function value than all the points on the function that are close to it.

A **relative minimum point** of a function is a point that has a value that is less than all the points on the function that are close to it.

EXAMPLE 1

The graph and table show the average cost of gasoline in the United States from 1918 to 2008 in 10-year intervals. The costs shown have been adjusted for inflation.



McMahon 2012

Year	1918	1928	1938	1948	1958	1968	1978	1988	1998	2008
Average cost	\$3.87	\$2.87	\$3.31	\$2.52	\$2.42	\$2.28	\$2.33	\$1.89	\$1.46	\$3.49

A From 1928 to 1978, what was the relative maximum cost of gasoline?

SOLUTION

The relative maximum cost of gasoline from 1928 to 1978 was \$3.31.

B From 1978 to 2008, what was the relative maximum cost of gasoline?

SOLUTION

The relative maximum cost of gasoline from 1978 to 2008 was \$3.49.

C From 1918 to 2008, what year had the relative minimum cost of gasoline?

SOLUTION

From 1918 to 2008, the relative minimum cost of gasoline was in 1998. ■

Determining Increasing, Decreasing, and Constant Intervals

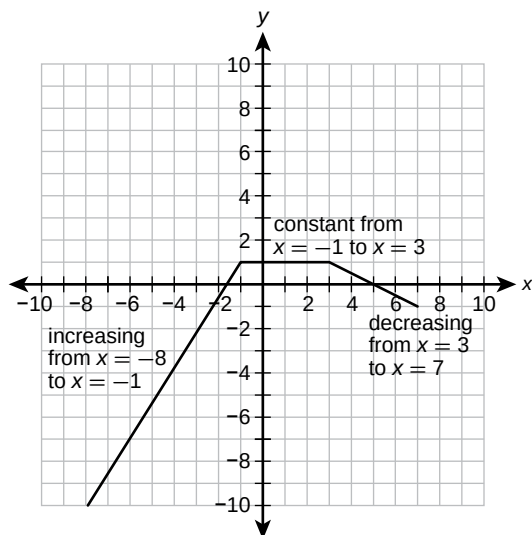
Definitions

A function is **increasing** within an interval if the value of $f(x)$ increases as the value of x increases.

A function is **decreasing** within an interval if the value of $f(x)$ decreases as the value of x increases.

A function is **constant** within an interval if the value of $f(x)$ remains the same.

The graph shows a function that changes from increasing to constant to decreasing between different intervals.

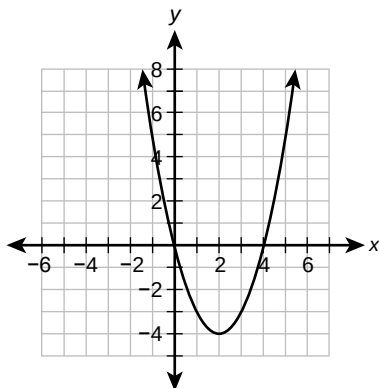


EXAMPLE 2

Consider the function $f(x) = x^2 - 4x$.

A Sketch the graph.

SOLUTION



B Determine the intervals for which the function is positive or negative.

SOLUTION

From the graph, you can see that the value of the function is positive (above the x -axis) when $x < 0$ and when $x > 4$. The value of the function is negative when x is between 0 and 4.

C Determine any relative maximum and minimum points.

SOLUTION

From the graph, the point $(2, -4)$ is a relative minimum point.

D Determine the intervals for which the function is increasing, decreasing, or constant.

SOLUTION

The function is decreasing when $x < 2$. The function is increasing when $x > 2$. ■

Classifying Functions as Even, Odd, or Neither

Definitions

A function $f(x)$ is **even** if $f(-x) = f(x)$ for all values of x . The graph of an even function is always symmetric about the y -axis.

A function $f(x)$ is **odd** if $f(-x) = -f(x)$ for all values of x . The graph of an odd function is always symmetric about the origin.

EXAMPLE 3

Determine whether the function is even, odd, or neither even nor odd.

A $f(x) = x^4 - 3x^2$

SOLUTION

Find $f(-x)$

$$\begin{aligned} f(-x) &= (-x)^4 - 3(-x)^2 \\ &= x^4 - 3x^2 \end{aligned}$$

The function $f(x) = x^4 - 3x^2$ is even because $f(-x) = f(x)$.

B $f(x) = \sin x$

SOLUTION

Find $f(-x)$.

$$\begin{aligned}f(-x) &= \sin(-x) \\ &= -\sin x\end{aligned}$$

The function $f(x) = \sin x$ is odd because $f(-x) = -f(x)$.

C $f(x) = 2x^3 - 3x^2 - 4x + 4$

SOLUTION

Find $f(-x)$.

$$\begin{aligned}f(-x) &= 2(-x)^3 - 3(-x)^2 - 4(-x) + 4 \\ &= -2x^3 - 3x^2 + 4x + 4\end{aligned}$$

The function $f(x) = 2x^3 - 3x^2 - 4x + 4$ is neither even nor odd because $f(-x) \neq f(x)$ and $f(-x) \neq -f(x)$. ■

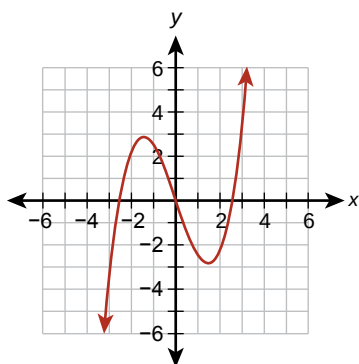
Identifying Even and Odd Graphs

You can determine whether a function is even, odd, or neither even nor odd by inspecting the graph. The graph of an even function is always symmetric about the y -axis, and the graph of an odd function is always symmetric about the origin. The graph of a function that is neither even nor odd is not symmetric about the origin or the y -axis.

EXAMPLE 4

Determine whether the function is even, odd, or neither even nor odd by examining the graph.

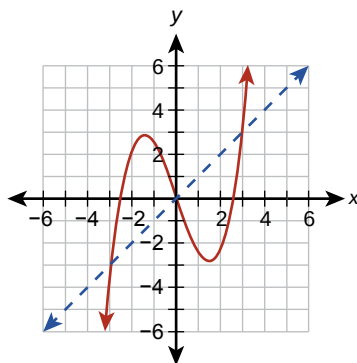
A

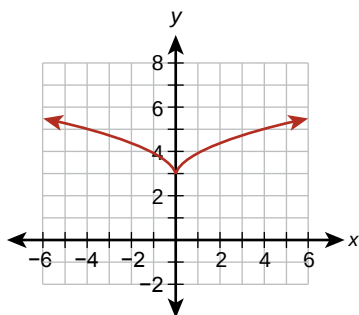


► **Think About It** If the point (x, y) is on the graph of an odd function, you will also find the point $(-x, -y)$ on the graph. If the point (x, y) is on the graph of an even function, you will also find the point $(-x, y)$ on the graph.

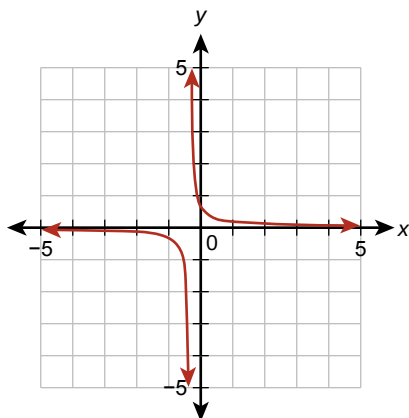
SOLUTION

The function is odd. The graph of an odd function is always symmetric about the origin. If you rotate the graph 180° about the origin, it is identical.



B**SOLUTION**

The function is even. The graph of an even function is always symmetric about the y -axis. If you fold the graph along the y -axis, it is identical.

C**SOLUTION**

The function is neither even nor odd. It is not symmetric about the origin or the y -axis. ■

Continuous and Discontinuous Functions

Definitions

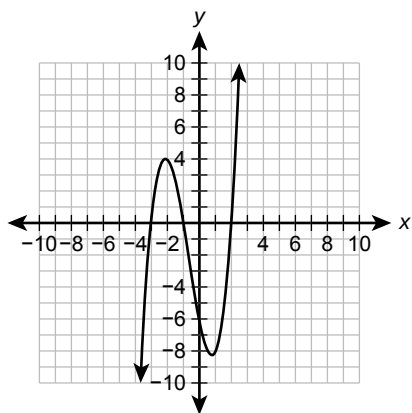
A function is **continuous** if every value of x within the domain of the function has a corresponding value for $f(x)$.

A function is **discontinuous** if there are values of x for which $f(x)$ is undefined.

EXAMPLE 5

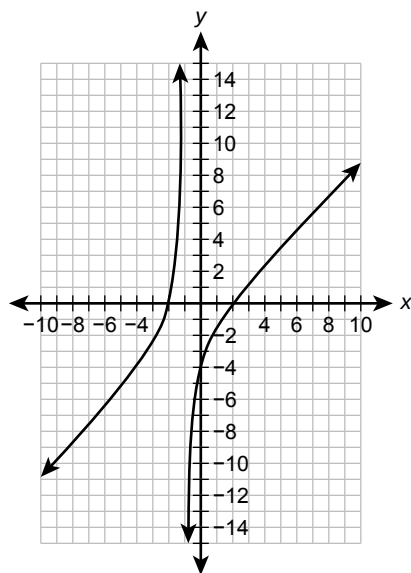
Determine whether the graph is continuous or discontinuous.

A

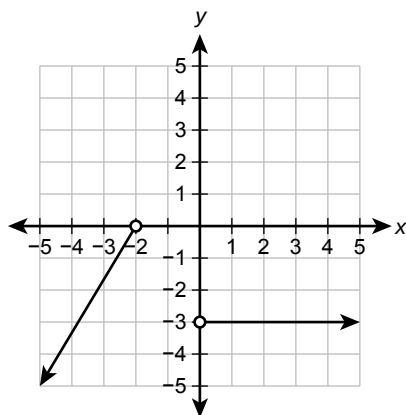


SOLUTION

This function is continuous over all real numbers. The graph of the function has no breaks, so every value of x has a corresponding value for $f(x)$.

B**SOLUTION**

This function is discontinuous over its domain. It has a vertical asymptote at $x = -1$, so the function is undefined for that point.

C**SOLUTION**

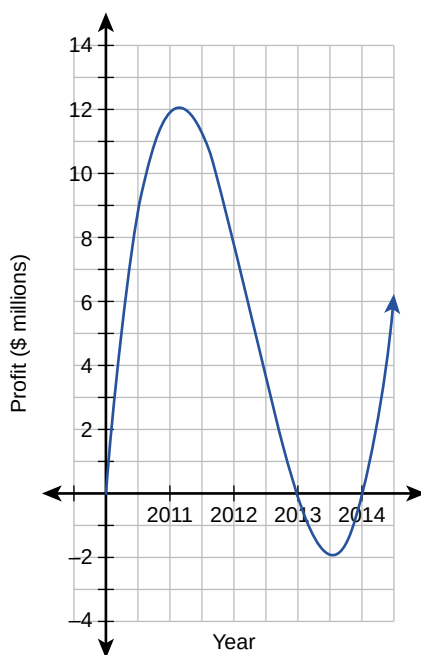
This function is discontinuous over its domain because it is not defined for $-2 \leq x \leq 0$. ■

Interpreting Real-World Polynomial Graphs

Real-world situations can be modeled with polynomial graphs. The features of polynomial graphs, including maximum or minimum points and x - and y -intercepts, can help interpret the data.

EXAMPLE 6

A software company tracked its profits each year from 2010 to 2014. The profits, in millions of dollars, are shown in the graph.



- A** Estimate the maximum and minimum points of the graph. What do they represent?

SOLUTION

The maximum point of the graph occurs at 2011, when the profit was \$12 million. The minimum point occurs between 2013 and 2014, when the profit was -\$2 million. These points represent the time periods when profits were highest and lowest.

- B** Locate the x - and y -intercepts and interpret what they mean in terms of profits.

SOLUTION

The x -intercepts are at $x = 2010$, $x = 2013$, and $x = 2014$, meaning that at the beginning of 2010, 2013, and 2014, the company's profits were at \$0. Since $x = 2010$ has a y -value of 0, it is also the y -intercept of the graph.

- C** During what time period were the company's profits negative?

SOLUTION

The company's profits were negative between 2013 and 2014 because the graph drops below the x -axis. ■